MEMORANDUM

TO	David / School Facilities CV Project
FROM	Rut Patel
DATE	September 3, 2026
SUBJECT	An evidence-ranked prototype for measuring school facilities from aerial imagery

Bottom line. On 66 date-compatible comparisons across nine audited schools, the frozen final pipeline was correct on 45 (68.2%) and mean confidence was 69.8%. The original seeded eight-school sample was 40/58 (69.0%); a ninth school was deliberately added as a solar-positive challenge and was 5/8. This is a small development audit, not an untouched test or national accuracy claim.

Fencing was the central failure: extent was correct for 3/9 schools and type for 2/9, causing 13 of 21 errors. All 13 fence errors received confidence 0.4 or 0.1. Excluding the two fence fields, accuracy was 40/48 (83.3%). The lesson is not to hide fencing, but to route weak overhead evidence to higher-resolution or street-level review.

Result	N	Observed rate	Meaning
All scored observations	66	68.2% correct	45 correct; mean confidence 69.8%
Seeded sample	58	69.0% correct	Original reproducible eight schools
Confidence = 0.9	31	30/31 (96.8%)	One high-confidence track miss
Calibration diagnostics	66	ECE 8.0 pp; Brier 0.103	Pooled mean gap alone can cancel errors
Confidence = 0.4 or 0.1	13	0/13 correct	Reviewing them leaves 45/53 correct
Field confidence <= 0.7	35	20/21 errors caught	Unreviewed fields are 30/31 correct

APPROACH

- I treated each CCD coordinate as a starting point, generated one Nominatim alternative, and manually opened map/aerial links for all 25 schools. I retained 24 CCD centers and corrected one.
- I downloaded dated NAIP imagery through Microsoft Planetary Computer. Each model request included the full campus, a centered detail crop, and four overlapping sections.
- Gemini 3.1 Flash-Lite returned a typed schema. Deterministic code then handled missing imagery, boundary ambiguity, unsupported fence/solar claims, confidence bands, and solar-area geometry.

VALIDATION METHOD

A fixed seed selected eight schools by level (3 primary, 2 middle, 3 high). I saved the initial predictions before labeling, then used three schools while improving the method. I later inspected mistakes in the other five during prompt/model testing; final predictions were produced after that work. Therefore these results audit the final artifact but are not an unseen test.

I manually added Spring Lake Heights after seeing rooftop solar so the audit included one date-compatible positive; it is marked as a targeted challenge, not a random row. I labeled only supported fields, made the final decisions myself, used ChatGPT/Claude only as second opinions for unfamiliar features, and left unresolved cells blank.

Categories/counts require exact agreement; an abstention counts wrong against a filled label; area requires <=25% relative error. Six filled references are retained but excluded from the primary score: four Bertha/Pittsburgh solar values compare newer/undated Google imagery to 2023/2022 NAIP, Spring Lake's area is a rough no-ruler estimate, and Cerra Vista's provisional seven-portable count uses undated Google imagery. I report every exclusion.

WHAT CALIBRATION SAYS

- 0.9: 30/31 correct (96.8%); 0.7: 15/22 correct (68.2%); 0.4/0.1: 0/13 correct.
- The bands rank risk well. Fixed-band expected calibration error is 8.0 points; the worst gap is 40 points in the five-item 0.4 band.

Overall confidence (69.8%) nearly matches accuracy (68.2%), but that pooled gap can cancel opposite band errors. Brier score is 0.103. Only nine schools contribute and fields within a school are related, so I do not fit or certify probabilities; a larger date-matched set is needed per field.

WHAT I WOULD FLAG NOW

- Review 0.4/0.1 fields: 19.7% workload, 13/21 errors caught, and 45/53 retained fields correct (84.9%).
- Review confidence <=0.7 when recall matters: 53.0% workload, 20/21 errors caught, and 30/31 retained fields correct (96.8%).
- Always review fence outputs, solar negatives from stale imagery, positive/area estimates, unclear boundaries, crop edges, shared facilities, and occluded counts.

The revised school flag includes unobservable fence type, catches 20/21 errors, and flags 88% of the full run. Field-level review is more useful because one weak fence need not send a clear pool back to a reviewer. These are same-audit trade-offs, not deployment guarantees.

FAILURES AND LIMITS

Square crops can include shared facilities. NAIP vintages vary; Ridgeview uses 2017 imagery. Thin fences are sub-pixel/occluded and Street View misses campus sides; houses, hedges, and roads mark edges but are not fences, so even manual fence labels are uncertain. All eight scored portable labels are zero, so 8/8 does not test positive detection; Cerra's roof seams may be seven buildings or joined modules. Solar area uses georeferenced boxes/fill fractions, not segmentation, and is capped at 0.4 pending a precise positive reference. Earlier same-model runs ranged 34-42/58; the frozen final is 40/58. Hosted runs also produced 503/429 errors, invalid/truncated responses, and latency outliers; freezing outputs preserves reproducibility.

HOW I WOULD SPEND 100 HOURS

Hours 1-20: learn and specify. Read work on facility mapping, solar detection, remote-sensing counts, and uncertainty calibration; compare imagery dates/resolution/licenses/APIs; inspect school, parcel, and building data; tighten the label guide; double-label a pilot to learn human disagreement.

Hours 21-40: campus boundaries. Test OSM/Overture buildings and land use, roads, open parcels, and district GIS layers. Build candidate polygons and review disagreements in a simple map UI.

Hours 41-65: real benchmark. Label 500-1,000 schools across geography, level, rurality, image age, and complexity; oversample rare positives; use two labelers plus adjudication; date-match sources and draw polygons/segments where area or counts matter.

Hours 66-90: attribute-specific methods. Compare VLMs with building/campus masks and specialized solar, portable, court, field, and track detectors. Treat fencing separately with higher-resolution aerial and a legally usable street-level source. Route only uncertain disagreements to a stronger model.

Hours 91-100: calibration and operations. Freeze a geographic test set; calibrate each field; choose review thresholds from error cost and staff capacity; then test drift, batching, caching, retries, lineage, latency, and cost together.

SCALE, COST, AND MODEL CHOICE

The final 25-school run used median 8.76 seconds and about 8,125 input plus 432 output tokens per school. At observed load, 130,000 schools cost about $348 with 3.1 Flash-Lite standard ($174 batch). The completed 3.5 test included thinking tokens and implies $5,404 standard ($2,702 batch); it gained only 1/58 correct fields, had slightly worse calibration, and a 93.7 s median. A 3.8 estimate is ~$1,000 if it used 3.1's token load, but repeated 503s prevent any accuracy claim. The free 500/day cap implies >=260 days; idealized 50-way runtime is 6.3 hours before limits/QA.

TIME AND AUTHORSHIP

Approximate time: 20 min understanding; 50 min approach/initial code; 30 min campus review; 60 min labeling the original eight; 90 min guardrails, prompt changes, fixes, and checks; 60 min model/final tests including the added solar case; 30 min memo/packaging. Total: 5 h 40 min. I knowingly used the extra 40 minutes to learn unfamiliar aerial features, resolve labels, and investigate failures rather than tune for a better-looking accuracy number.

DATA, MODEL, AND SOFTWARE

CCD supplied starting coordinates; NAIP came from Microsoft Planetary Computer STAC; Nominatim provided cached alternatives; Google Maps/Street View was manual validation only; Gemini 3.1 Flash-Lite performed structured extraction. Python, pandas, rasterio, pyproj, Pillow, Pydantic, and pytest handled the pipeline. ChatGPT helped with initial design/code and visual second opinions; Claude with visual second opinions; Codex with code review, tests, prompt revisions, and packaging.

I read and reviewed every assistant-generated code or prompt change, traced how it affected the data flow, ran it, inspected outputs, and revised or rejected suggestions that did not fit the task. I used AI side by side, not as ground truth; definitions, campus decisions, labels, experiments, and reporting are mine.

SOURCES

[1] Microsoft Planetary Computer, NAIP dataset: planetarycomputer.microsoft.com/dataset/naip
[2] OSM Foundation, Nominatim Usage Policy: operations.osmfoundation.org/policies/nominatim/
[3] Overture Maps documentation: docs.overturemaps.org/
[4] Google AI, Gemini models and pricing: ai.google.dev/gemini-api/docs/models and /pricing